



DOW™ Electrical & Telecommunications HFDG-4201 NT Crosslinkable Power Cable Insulation Compound

Overview

HFDG-4201 is a low-density, crosslinkable, unfilled polyethylene compound designed for use in power cable applications of 10 KV or less.

Specifications

Cables insulated with HFDG-4201, using sound commercial fabrication practice, should meet the latest editions of the following industry cable specifications:

- ICEA: S-66-524/NEMA WC7
- IEC: 60502
- GB: 12706

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.920 g/cm ³	0.920 g/cm ³	ASTM D792
Degree of Crosslinking - Extractables	< 20 %	< 20 %	ASTM D2765A
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength	2800 psi	19.3 MPa	ASTM D638
Tensile Elongation (Break)	500 %	500 %	ASTM D638
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Hot Creep	< 100 %	< 100 %	ICEA T-28-562
Hot Set	< 10 %	< 10 %	ICEA T-28-562
Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Volume Resistivity	1.0E+15 ohms·cm	1.0E+15 ohms·cm	ASTM D257
Dielectric Strength			ASTM D149
0.125 in (3.18 mm) ¹	660 V/mil	26 kV/mm	
0.125 in (3.18 mm) ²	590 V/mil	23 kV/mm	
Dielectric Constant (60 Hz)	2.30	2.30	ASTM D150
Dissipation Factor (60 Hz)	5.0E-4	5.0E-4	ASTM D150

Additional Information

Nominal property values above represent tests on molded stress-relieved slabs. Cure times were 15 minutes at 347°F (175°C).

Extrusion	Nominal Value (English)	Nominal Value (SI)
Melt Temperature	240 to 280 °F	116 to 138 °C

Extrusion Notes

HFDG-4201 provides excellent surface finish and outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures in the range of 240 to 280°F (116-138°C) are recommended, although higher melt temperatures are possible on certain equipment with due care. Specific recommendations for processing conditions can be determined for the application and type of processing equipment in use.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Method A (Short-Time)

² Method B (Step-by-Step)

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Published: 2005-11-17

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